

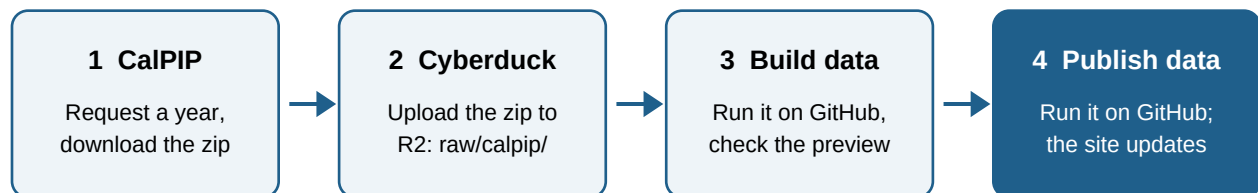
People and Pesticide Explorer: Updating the Data

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This guide replaces the March 2025 version. The explorer's data moved from Amazon Web Services (AWS) to Cloudflare R2 and GitHub Actions. Requesting data from CalPIP hasn't changed.

Adding a year of pesticide use data takes four steps:

- 1 Request the year from CalPIP and download the zip file. This takes about 20 minutes, mostly waiting for an email.
- 2 Upload the zip to the project's storage on Cloudflare R2, using the Cyberduck app.
- 3 Run the **Build data** job on GitHub and check the results. This takes about 5 minutes.
- 4 Run the **Publish data** job to put the new data on the live site.



Nothing changes on the live site until step 4, and step 4 can be undone at any time.

Until the switch-over. The explorer is moving to this new data system. Until Dylan confirms the switch is done, Publish data doesn't change what the live explorer shows, so check with Dylan before updating data.

Before you start

- **An upload key** for the project's storage: three values (Server, Access Key ID, Secret Access Key) that Dylan sends you. Keep them private, like a password.
- **Cyberduck**, a free app for Mac and Windows: cyberduck.io.
- **A GitHub account** with permission to run Actions in the **open-spatial-lab/cpr** repository. Dylan can add you.

Storage bucket: pesticide-data

Explorer website: _____

Key terms

- **CalPIP**: the website of the California Department of Pesticide Regulation (DPR) for downloading Pesticide Use Report (PUR) data.
- **Cloudflare R2**: online file storage, similar to Dropbox or Google Drive. The explorer's data lives here.
- **Bucket**: the top-level folder for a project in R2. This project's is `pesticide-data`.
- **Cyberduck**: a free app for moving files to and from online storage such as R2.
- **GitHub Actions**: buttons on GitHub that run scripts. *Build data* and *Publish data* are two of them.
- **Build**: a complete processed copy of the explorer's data, named by the date and time it was made, such as `v20270115-0930`. Builds are never changed or deleted, which is what makes undoing possible.
- **ZIP archive**: a compressed file, CalPIP's download format. Upload it as it is.

1. Request data from CalPIP

CalPIP (calpip.cdpr.ca.gov) is DPR's portal for pesticide use data. The explorer needs the full record-level data for the whole state: one row per pesticide application, not a summary. CalPIP allows one year per request, so repeat these steps for each year you're adding.

- 1 Open CalPIP and choose **PUR** as the data source. Click **Year** in the navigation menu and select the year you want. Leave the location, site, product and chemical filters empty.

The screenshot displays the CalPIP application interface. At the top, there is a navigation bar with links for 'About DPR', 'Join E-Lists', 'Careers', 'Contact Us', and a 'Select Language' dropdown. Below this is the 'California Department of Pesticide Regulation' logo and a search bar. The main content area is titled 'CALIFORNIA PESTICIDE INFORMATION PORTAL (CALPIP) APPLICATION' and contains a 'Query Pesticide Use Report' section. This section has a 'Select a Year' dropdown menu with a list of 'Available Years (32):' from 2022 down to 2017. The 'Year Selected' is 2022. To the left of the main content is a 'CalPIP Navigation Menu' with various options: 'CalPIP Application Home', 'My Selections', 'Data Sources' (PUR, GWPA, PRESCRIBE), 'Selected Data Source' (Pesticide Use Report), 'Search by Data Category' (Date, Location, Site/Crop, Product, Chemical), and 'Chemical'. The 'Date' category is expanded, showing 'Year' as the selected option. At the bottom of the 'Query Pesticide Use Report' section, there are two buttons: 'My Selections' and 'Start Over'.

- Click **Format Output** in the navigation menu. Make sure **Summarize the Data** is **not** checked, and keep all output columns selected. The build checks for the columns it needs and will say if any are missing.

The screenshot shows the top navigation bar of the CalPIP web application. The header includes the CA.GOV logo, social media icons, and links for About DPR, Join E-Lists, Careers, and Contact Us. A language selection dropdown is set to 'Select Language'. Below the header, the California Department of Pesticide Regulation (dpr) logo is displayed. The main navigation menu includes links for About, Pesticide Safety, SPM and IPM, Pesticide Registration, Professional Licensing, Data and Reports, Get Involved with DPR, and Search. The 'Summarize Data' section contains a checkbox for 'Summarize the Data' which is unchecked. Below this, there is explanatory text about summarizing data and a 'Sort-By Columns' section. The 'Sort-By Columns' section includes a search bar, 'Add All', 'Add Selected', 'Search', and 'Clear Search' buttons. A message indicates '35 items displayed'.

- Under **Output File Format/Type**, keep **Tab-delimited text (default)** checked and **HTML table** unchecked. Click **Submit Query**.

The screenshot shows the 'Output File Format/Type' section of the CalPIP web application. The top navigation bar and header are identical to the previous screenshot. The main navigation menu is also the same. Below the header, the 'Output File Format/Type' section is displayed. It includes a table with 4 items selected: COUNTY_NAME, SITE_NAME, PRODUCT_NAME, and CHEMICAL_NAME. Above the table are 'Remove All' and 'Remove Selected' buttons. Below the table, there is a confirmation message and a section for 'Output File Format/Type'. This section has two checkboxes: 'Tab-delimited text (default)' which is checked, and 'HTML table' which is unchecked. Below this, there is explanatory text about file formats and a message: 'To continue selecting query criteria, click on another category in the CalPIP Navigation Menu.' At the bottom of the section are four buttons: 'Reset Page', 'My Selections', 'Start Over', and 'Submit Query'. The footer of the page indicates 'Version 2024.01 (2022 PUR Data Update)'.

- 4 Enter your email address and click **Send Query Now**.

The screenshot shows the CalPIP web application interface. At the top is a blue header with the CA.gov logo, social media icons, and navigation links like 'About DPR', 'Join E-Lists', 'Careers', 'Contact Us', and a 'Select Language' dropdown. Below this is a white navigation bar with the 'dpr' logo and links for 'About Pesticide Safety', 'SPM and IPM', 'Pesticide Registration', 'Professional Licensing', 'Data and Reports', and 'Get Involved with DPR'. The main content area is divided into two columns. The left column, titled 'CalPIP Navigation Menu', contains links for 'CalPIP Application Home', 'My Selections', 'Data Sources' (with sub-links for PUR, GWPA, and PRESCRIBE), 'Selected Data Source' (Pesticide Use Report), 'Search by Data Category' (with sub-links for Date, Year, Location, County, COMTRS, Zip Code, Site/Crop, Name Search, Product, Name Search, Chemical, Name Search, and Other Criteria). The right column, titled 'Query Pesticide Use Report', contains instructions: 'Your query is ready to be sent to the database to extract data. You will be prompted for your email address if you have not entered it below. When your query has been run you will be notified by email with the URL to download the data file.' It features an 'Enter your email:' text input field with a '(REQUIRED)' label below it. A red warning message states: 'Please review your email address for accuracy. If your address is entered incorrectly, we will be unable to notify you when your data results are ready.' Below this, it says 'You will be contacted through the email address you have entered strictly regarding your CalPIP query and not for any other purpose. You are responsible for picking up your data file. YOUR DATA FILE WILL BE DELETED IN 7 DAYS WHETHER IT IS CLAIMED OR NOT.' There is a link 'Click here to request CalPIP data:' and a 'Send Query Now' button. At the bottom of the right column, it says 'Click 'My Selections' to see a summary of all of the selections you have made. Click 'Start Over' to clear all CalPIP selections.' and includes 'My Selections' and 'Start Over' buttons.

- 5 CalPIP emails you a download link, usually within about 20 minutes. Download the zip file. Don't unzip it, rename the file inside it, or edit it. The link expires after 7 days.

2. Upload the zip with Cyberduck

The zip goes in the folder `raw/calpip/` in the `pesticide-data` bucket. That folder holds one file per year: zips that people have uploaded, and files named `calpip_<year>.parquet` for older years. Don't delete or rename those; every build uses all of them. Cyberduck handles files of any size.

Connect (first time only)

- 1 Open Cyberduck and click **Open Connection**. In the list at the top of the window, choose **Amazon S3**.
- 2 Enter the **Server**, **Access Key ID** and **Secret Access Key** that Dylan sent you, exactly as sent. Click **Connect**.
- 3 To skip these steps next time, save the connection: choose **Bookmark**, then **New Bookmark**. After that, double-click the bookmark to connect.

Upload

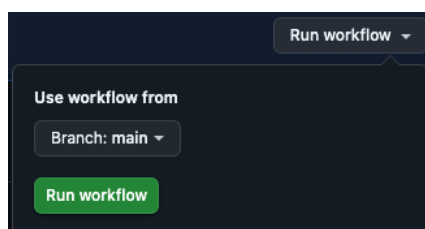
- 1 Double-click `pesticide-data`, then `raw`, then `calpip`.
- 2 Drag the zip from your computer into the Cyberduck window. Wait until the transfer window says it's complete.

Stay in raw/calpip/. The upload key can change anything in the bucket, including the data the live site uses. Only add or delete files in raw/calpip/.

Replacing a year. DPR sometimes revises past data. To re-pull a year, request it from CalPIP again, upload the new zip, and delete the old zip for that year if there is one (select it in Cyberduck and press Delete). If both are in the folder, the build uses the one uploaded most recently.

3. Build the data

- 1 Open [the Build data page](#) (in the repository, click **Actions**, then **Build data** in the list on the left).
- 2 Click **Run workflow**, then the green **Run workflow** button.



- 3 Wait for the run to finish, usually about 5 minutes. Refresh the page: a green check means it worked, a red X means it stopped (see Troubleshooting).
- 4 Click the run to open its summary page and check these things:
 - **The version name and year range**, such as v20270115-0930, years 2017–2024. The year you added should be the last year. Copy the version name; you'll need it to publish.
 - **The yearly totals table.** The year you added shows as *new year*. Years already on the site should show 0.00% unless you re-pulled that year. A large change in an older year is a reason to stop and ask before publishing.
 - **Chemicals with no class or health information**, if the summary lists any. They still count in totals, but the Chemical Class, Use Type and Health filters skip them until PAN's category spreadsheet (AI Cat Data.xlsx) has them. You can still publish; send the list to whoever keeps that spreadsheet.
 - **The preview link.** It opens the explorer with this build's data and a banner naming the version you're previewing. Pick the new year in the date range and try a map and a chart.

Building doesn't change the live site. You can run Build data as many times as you like; the preview link always shows the most recent build.

4. Publish the data

- 1 Open [the Publish data page](#) (**Actions**, then **Publish data**).

- 2 Click **Run workflow**. In the **version** box, paste the version name from the Build data summary. If you leave it empty, it publishes the most recent build. Click the green **Run workflow** button.
- 3 It finishes in under a minute. Reload the explorer and it shows the new data. The website itself doesn't need to be rebuilt.

Undoing a publish. The Publish data summary page names the version it replaced. To go back, run Publish data again and type that version name into the **version** box. Every build is kept, so you can return to any of them.

Troubleshooting

Problem	What to do
Build data shows a red X, and the error mentions <i>missing CalPIP columns</i> .	The zip is a summary or is missing columns. Request the year again (section 1, steps 2 and 3), replace the zip, and run Build data again.
The year you added isn't in the totals table.	The zip isn't in <code>raw/calpip/</code> , or it was unzipped before uploading. Check the folder and that the file name ends in <code>.zip</code> .
Build data fails with a different error.	Send Dylan the link to the failed run. The live site isn't affected.
The CalPIP download link has expired.	Submit the CalPIP query again.
Cyberduck can't connect, or says <i>Access Denied</i> .	Check that the connection type is Amazon S3 and that the three values from Dylan were entered exactly, with no extra spaces. If it still fails, ask Dylan to check the key.
The site shows the wrong data after publishing.	Run Publish data with the previous version (section 4, Undoing a publish).

For maintainers: setup and operations are in `RUNBOOK.md` in the repository.